Doppler-Aided Link Adaptation for Underwater Acoustic Links

Main idea, gaps, methodology and theoretical objectives — for group discussion

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8 October 2026 (revised 9 October 2026 after the prior-art search)

Target: IEEE JSAC special issue on Next-Generation Underwater Acoustic Communication Systems (deadline 31 March 2027)

CLAIM stated in the idea document, not yet established SUGGESTED design option added after review OPEN needs material, a proof or an experiment SOURCE found by targeted search, not yet checked

Main idea

The proposal in one slide

Every packet, data or ACK, carries a free measurement of the Doppler scale $\epsilon = v_{\mathrm{rel}}/c$, the same at both ends. Both ends fuse them (and motion sensors, if any) into one kinematic forecast of the drift $d = \dot{\epsilon}$ and its uncertainty, which drives four blocks:

- **frame**: extra pilots where the forecast is uncertain, over a coherence-limited grid; packet length, modulation
- **receiver**: pre-warped along the forecast, tracking only its error
- **transmitter**: bulk Doppler pre-compensated, so several nodes can share the uplink at once
- **decision**: threshold rule on the predicted PER, margin from delayed ACKs; uplink guard bands and admission from σ_e

Candidate claim SUGGESTED: Only a forecast, learned from the link's own two-way Doppler, lets a transmitter pre-compensate, so several nodes share an uplink (NTN: from ephemeris). Bandit: follow-up.

Scope

Delay-tolerant traffic
Mobile or surface-coupled nodes
Seas with a swell component
Paths resolvable in delay or Doppler
One link, or K nodes to a gateway ($K = 1$: one link)

Still, fixed links: fall back to the baseline; "no harm" once overheads are counted is still to be shown.

Gaps with the literature

Related work (1): adapting the link despite stale feedback

Line of work	Representative papers	Difference
Physics-informed, uncertainty-aware AMC (closest)	Anjangi et al., UComms 2018 (Doppler spread as AMC feature); Wu, Chitre, Anjangi, JOE 2025 (GP interval BER + feedback scheduling)	MCS only; no kinematic forecast, packet layout or timing
Channel prediction for AMC	Radosevic et al., OCEANS 2011, JOE 2014; Tian et al., OCEANS 2026; Wang, OCEANS 2022; Jing, IoT-J 2025 (delay-Doppler)	Predict the channel, not the motion; MCS or bit loading
Prediction instead of feedback	Park et al., Remote Sens. 2026 (closest to our decision rule)	Doppler index, margin, PER target; Raptor overhead only
Feedback timing	AoI-aware hierarchical bandit (Busacca et al., GLOBECOM 2025)	No physical model
Error control under delay	OLLA (LTE/5G); ACI (Gibbs–Candès 2021); delayed ACI (El Halabi–Brandt 2026)	Same update rule as our margin; new: action-dependent delay, service accounting
AMC survey and benchmark	Cooper-Baldock et al., Appl. Ocean Res. 2026 (open-data benchmark)	Baselines to compare against

Details: `literature/prior_art_summary.md`.

Related work (2): Doppler is compensated, and packets are designed offline

Line of work	Representative papers	Use of Doppler
Physics of wave-induced Doppler	Walstead, JASA 2016–17; Rawat, IEEE TAES 2021 (Doppler spectrum from sea-surface motion); Li et al., JOE 2008 (residual offset varies with waves)	Characterisation; design of compensation
Packet structure for a motion range	Zakharov et al., JOE 2019 ($\lvert a\rvert \leq 0.25\,\mathrm{m/s^2}$ design, ≤ 1 simulated; $\approx$1 s packets)	Fixed offline design, not adapted online
Acceleration-aware compensation	Abdelkareem et al., Ad Hoc Netw. 2016 (previous packet gates the search); Cho–Kim, IET Commun. 2026; Hu et al., MSN 2024 (Kalman)	Absorbs constant drift; models stay inside one packet
Doppler prediction in the receiver	Ceballos Carrascosa and Stojanovic, JOE 2010 (Doppler predicted across OFDM blocks)	In the receiver; not used by the transmitter
Doppler of both ends	DE-Sync 2018, APE-Sync 2019 (returned in the reply); Aubry 2019, Choi 2022 (navigation)	Clock sync, position; not the link

Enablers now public: Underwater Acoustic Channel Library (Li et al., 2026; bulk drift kept, one-way, $\approx$50–60 s records), Qarabaqi–Stojanovic simulator, NOAA NDBC wave spectra. Caution: Watermark may remove bulk Doppler.

Related work (3): the closest analogues are outside underwater acoustics

Work	Shared mechanism	What it lacks
Yuan et al., JSTSP 2021 (V2I ISAC)	Prediction drives TX, pilots and RX: the concept	Constant velocity; no two-way Doppler, PER layer
Simeone–Spagnolini, ICC 2004	Pilots from Kalman covariance	Fading, not kinematics
Li et al., arXiv 2606.16717	Wave-driven buoy state model	RF beamforming
3GPP NTN (TS 38.300)	Continuous Doppler pre-compensation	External ephemeris
DE-Sync 2018, APE-Sync 2019	Far end's Doppler in the reply	Clock skew only

The architecture is not new in general: claim the underwater instance and its analysis (R1–R4).

OPEN Scholar cited-by of Yuan 2021, DE-Sync, APE-Sync not yet checked.

Related work (4): multi-user Doppler is handled at the receiver

Work	Mechanism	What it lacks
Daoud et al., IEEE TVT 2015	Multiple resampling at the receiver, one branch per user	No prior; receiver only
Tu, Duman, Stojanovic, Proakis 2011, 2013	OFDMA; one resampling branch per user at the receiver	K branches; interference remains
Zhang et al., arXiv 2026	Concurrent random access; per-user power control	Path-specific Doppler left in the receiver
Hasan et al., Ad Hoc Netw. 2025	Guard bands adapted to mobility (metaheuristic)	No forecast, no pre-compensation
Oiwa et al., IEICE Tech. Rep. 2020	Accelerometer-aided Doppler compensation	Own node only; not peer reviewed
3GPP NTN (TS 38.300)	UE pre-compensates Doppler before sending	Ephemeris, not learned from the link

Library, web, Scholar and 159 citing works checked (9 Oct 2026): no UWA transmitter pre-compensates its own Doppler. Say "transmit-side": UWA papers use "Doppler pre-compensation" for a receiver step.

Candidate gaps, re-ranked after the library check

G1 No packet places training from the error variance of a kinematic Doppler forecast

Pilots follow mobility *statistics* (NR DMRS, 802.11bd) or spreads (Mangione 2021); Kalman-covariance pilots: radio only (Simeone–Spagnolini 2004).

G2 No receiver carries a Doppler forecast with uncertainty across packets

Kinematic models stay inside one packet (Cho–Kim 2026, Hu 2024); the previous packet only gates the search (Abdelkareem 2016).

G3 Doppler from both ends is not tracked as a kinematic process for the link

Exchanged for clock sync (DE-Sync, APE-Sync); fed one way to navigation (Aubry 2019, Choi 2022).

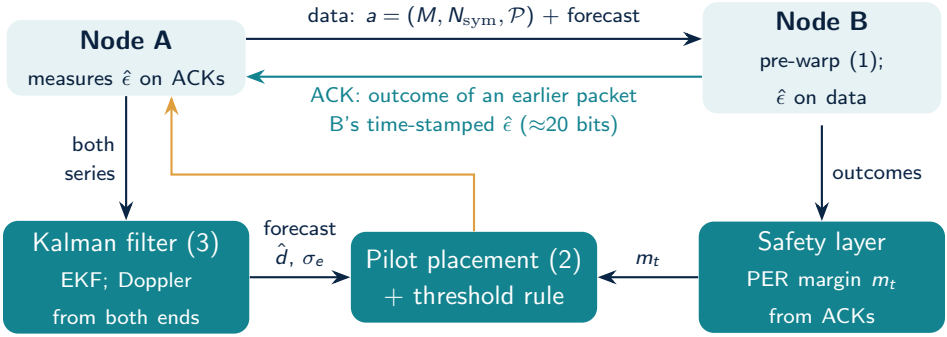
G5 No UWA transmitter pre-compensates Doppler for a multi-user uplink

Per-user Doppler is separated at the receiver, one resampling branch per user (Tu et al. 2011, 2013); guard bands follow mobility (Hasan 2025).

G4 (learning with arm-dependent delay): follow-up paper. Dropped: send timing to the swell.

Proposed methodology

System at a glance



Shown for one link ($K = 1$); numbers in brackets: blocks on the following slides; the uplink with K nodes adds Block 4. OPEN For the technical specification: header fields and quantisation; measurement ordering (out of sequence only with pipelined packets); SNR source; NACK or timeout, packet matching, ACK vs data loss; whether $\hat{\epsilon}$ needs payload decoding or only preamble acquisition.

Two quantities that must not be confused

Doppler scale (dimensionless)

$$\epsilon(t) = \frac{v_{\mathrm{rel}}(t)}{c}$$

Physical Doppler drift (units s^{-1})

$$d(t) = \dot{\epsilon}(t) = \frac{\dot{v}_{\mathrm{rel}}(t)}{c}$$

Some papers call ϵ the "Doppler rate". Here d is always the *physical* drift, never the receiver's residual.

- What harms a packet is what remains after compensation: it depends on receiver and action, not on d alone: forecast error $e \to$ receiver, action $a \to$ residual impairment $\phi_a(e) \to$ PER
- Example: $|d| = 10^{-4}\,\mathrm{s}^{-1}$ is a relative acceleration of $0.15\,\mathrm{m/s}^2$ at $c = 1500\,\mathrm{m/s}$.
- One common (bulk) component is an initial restriction. If path powers, delay spread, differential Doppler or tracking state change PER at equal d, that is a model limitation OPEN.

Model

Motion CLAIM: $v_{\mathrm{rel}}(t)$ = slow platform drift + damped oscillators (swell, wind sea) learned online.

Physical drift $d = \dot{v}_{\mathrm{rel}}/c$ in s^{-1}, common (bulk) component only: an initial restriction OPEN.

Observations: time-stamped $\hat{\epsilon}$ on the preamble of every packet received at *either* end (accuracy set by preamble length and bandwidth); ACK or timeout for earlier packets.

Forecast error $e(u) = d(u) - \hat{d}(u)$ over the receive interval, standard deviation $\sigma_e(u)$ from the estimator. A receiver without the prior has $\hat{d} = 0$, so $e = d$ (the baseline).

Failure CLAIM: the receiver turns e into a residual impairment $\phi_a(e) \geq 0$, depending on $(M, N_{\mathrm{sym}}, \mathcal{P})$; a monotone scalar curve is a hypothesis, checked by calibrating each receiver in simulation:

$$\mathrm{SNR}_{\mathrm{eff}} = \frac{\mathrm{SNR}}{1 + \mathrm{SNR}\,\phi_a(e)} < \gamma_a \iff \phi_a(e) > \frac{1}{\gamma_a} - \frac{1}{\mathrm{SNR}}, \qquad e_{\mathrm{tol}}(a, \mathrm{SNR}) = \sup\left\{|e| : \phi_a(|e|) \leq \tfrac{1}{\gamma_a} - \tfrac{1}{\mathrm{SNR}}\right\}$$

Empty set (e.g. $\mathrm{SNR} < \gamma_a$): a excluded; unbounded: $e_{\mathrm{tol}} = \infty$; non-monotone ϕ_a: use the set itself.
γ_a: SNR at which a meets the PER target, a measured threshold SUGGESTED.

Traffic: bursty arrivals; waiting $h \leq h_{\max}$ set by the MAC; a delay budget per packet.

A minimal action space SUGGESTED

$$a = (M,\ N_{\mathrm{sym}},\ \mathcal{P})$$

- M: modulation order, QPSK or 16-QAM
- N_{sym}: OFDM symbols in the packet
- $\mathcal{P} \subset \{1, \ldots, N_{\mathrm{sym}}\}$: positions of full training symbols, $N_{\mathrm{tr}} = |\mathcal{P}|$

The receiver pre-warp is not an action: it always follows the forecast. Fixed in the first paper: useful symbol duration T_s, cyclic prefix T_{cp}, code. Waiting time h: set by the MAC.

Packet duration

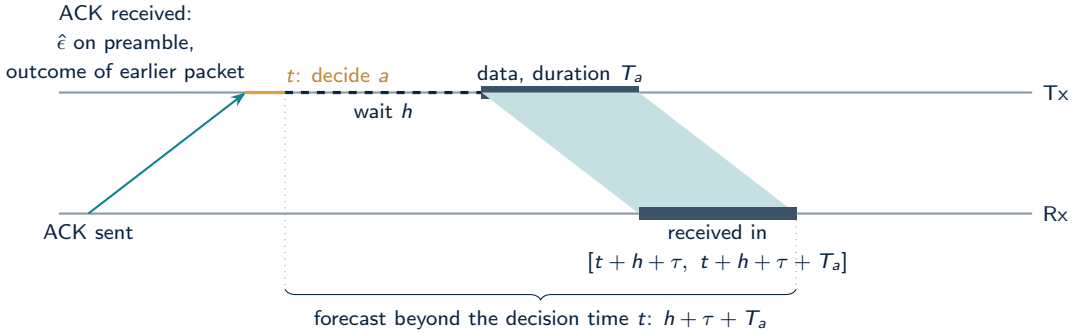
$$T_a = T_{\mathrm{pre}} + N_{\mathrm{sym}}\,(T_s + T_{cp})$$

Payload

$$B_a = \left(N_{\mathrm{sym}} - |\mathcal{P}|\right) N_{\mathrm{sc}}\, \log_2 M\, R_c$$

Uniform $\mathcal{P}$ with spacing K_{upd} is the baseline. Exact count from the frame (first estimate from the preamble? final training symbol?) OPEN. B_a: information bits before header, CRC, termination; split messages count every segment and its delay.

When the packet is received, and what is known then



τ: one-way propagation time ($\approx$0.67 s/km). t_{obs}: epoch of the physical state that $\hat{\epsilon}$ represents, fixed by the measurement equation OPEN. From an estimate at t_{obs} the horizon is $H = (t - t_{\mathrm{obs}}) + h + \tau + T_a$; if the filter is already propagated to t, $h + \tau + T_a$ remains. The ACK of this packet arrives $\geq h + 2\tau + T_a$ after the decision, so the margin reacts to old outcomes.

Shorter packets or more frequent tracking?

Doppler perfectly corrected at the start of an interval, then held fixed; roughly constant drift, narrowband around f_c:

$$|\Delta f(T)| \approx f_c\,|d|\,T \quad \Rightarrow \quad T \leq \frac{\Delta f_{\max}}{f_c\,|d|}$$

T is the interval without a compensation update:

- blockwise receiver: $T =$ whole packet $\Rightarrow$ shorten N_{sym}
- tracking receiver: $T = T_{\mathrm{upd}} = K_{\mathrm{upd}}(T_s + T_{cp})$ $\Rightarrow$ reduce K_{upd}, keep N_{sym}
- forecast-aided (Block 1): $|d| \to |d - \hat{d}|$

Illustration only SUGGESTED

$f_c = 20\,\mathrm{kHz}$, assumed $\Delta f_{\max} = 2\,\mathrm{Hz}$:
$|d| = 10^{-4}\,\mathrm{s}^{-1} \Rightarrow T \leq 1\,\mathrm{s}$
$|d| = 2 \times 10^{-4}\,\mathrm{s}^{-1} \Rightarrow T \leq 0.5\,\mathrm{s}$

$\Delta f_{\max}$ is not a measured receiver threshold.

Left out: wideband resampling effects, differential path Doppler, accumulated phase error, estimator noise, the actual detector. Valid only for fixed correction and roughly constant drift: it explains the mechanism, it is not a reliability guarantee.

Block 1: receiver pre-warped along the forecast SUGGESTED

Motion time-warps the received signal: $r(t) \approx s\big(t + \int_0^t \epsilon(u)\,du\big)$.

Today: estimate ϵ on the preamble (constant, or constant + acceleration), resample, track.

Here: resample along the forecast $\hat{\epsilon}(t)$ over the whole packet; the tracker sees only the residual warp $\int(\epsilon - \hat{\epsilon})$:

$$|\Delta f(T)| \approx f_c\,|d - \hat{d}|\,T \quad \Rightarrow \quad T \le \frac{\Delta f_{\max}}{f_c\,|d - \hat{d}|}$$

Same simplified model as before, with the drift replaced by the forecast error. Closest: Ceballos Carrascosa–Stojanovic, JOE 2010; Abdelkareem 2016 (gating only).

What it buys

- fewer pilots at equal PER, or longer packets, even if acceleration-aware receivers absorb the drift
- PER now depends on the forecast uncertainty σ_e, which the estimator reports

What it needs OPEN

- the forecast at node B: a few header bits, or the same filter running there
- a fallback if the forecast is wrong: residual test, preamble-only

“More pilots” made explicit

Five different things are called pilots:

1. more frequent temporal tracking observations
2. more pilot subcarriers inside one OFDM symbol
3. periodic full training symbols or midambles
4. known preambles at packet boundaries
5. decision-directed observations, if the receiver supports them

Against *temporal* drift the variable that matters is the spacing between usable tracking observations, K_{upd}.

Baseline: uniform spacing $K_{\mathrm{upd}} \in \{8, 4, 2\}$, test points, not thresholds. Proposed: a uniform grid at the largest spacing K_{max} that channel estimation allows, plus extra positions from the forecast uncertainty (Block 2) SUGGESTED. Per-symbol tracking needs embedded pilots and its own overhead model OPEN.

An observation and a receiver update are not the same operation: each receiver must state its estimator and what it needs from an observation OPEN.

Block 2: pilots where the forecast is uncertain SUGGESTED

The estimator predicts how the error variance $\sigma_e^2(u)$ grows over the packet, $u \in [0, T_a]$; each training symbol resets it locally.

Placement problem

$$\min_{\mathcal{P} \supseteq \mathcal{P}_0:\ |\mathcal{P}| \le N_{\mathrm{tr}}} \int_0^{T_a} \sigma^2_{e|\mathcal{P}}(u)\, du$$

$\mathcal{P}_0$: uniform grid at the largest spacing $K_{\max}$ the tap coherence time allows, so equalisation never loses the channel; the forecast adds only the extra pilots. Baseline: $\mathcal{P} = \mathcal{P}_0$.

OPEN $K_{\max}$ is receiver-dependent (comb pilots relax it; $K_{\max} \ge N_{\mathrm{sym}}$: $\mathcal{P}_0 = \emptyset$); integral or max over u; not submodular in general (Jawaid–Smith 2015).

Expected structure CLAIM (to derive)
For a swell $d = A\sin(\omega t + \varphi)$, a phase error $\delta\varphi$ gives a drift error $A\,\delta\varphi\cos(\omega t + \varphi)$: largest where d crosses zero, where the jerk peaks. Phase error grows with the horizon, so it dominates.

$\Rightarrow$ extra pilots cluster around the zero crossings of d. The jerk, which hurts acceleration-aware receivers most, decides where the extra pilots go.

Theory target (R3): closed-form placement for one swell component; weak-submodularity bound in general; overhead saved vs uniform spacing (see Risks).

Block 3: both ends measure Doppler SUGGESTED

Node B measures $\hat{\epsilon}$ on each data packet, node A on each ACK. Both see the same range rate, so a Kalman filter on the drift-plus-oscillators model fuses the two interleaved series (an EKF while the sea parameters are learned) SUGGESTED.

Largest gap between samples (stop-and-wait):

$$\Delta_{\text{one-way}} \geq 2\tau + T_a, \qquad \Delta_{\text{two-way}} \approx \tau + T_a$$

No aliasing for a swell of period P ($\Delta < P/2$):

$$R < \tfrac{c}{2}\left(\tfrac{P}{2} - T_a\right) \;\longrightarrow\; R < c\left(\tfrac{P}{2} - T_a\right)$$

$P = 8\,\text{s}$, $T_a = 1\,\text{s}$: $2.25\,\text{km} \rightarrow 4.5\,\text{km}$.

Cost and open points OPEN

- B's estimate returns in the ACK header: value + time stamp, ≈20 bits SUGGESTED
- it arrives with A's own sample and is older: processed in time order; out-of-sequence only with pipelined packets SUGGESTED
- holds both ends' motion projected on the path, unlike one IMU
- each direction may lock onto a different path: next slide

Theory target (R2): observability and forecast error vs range, one- vs two-way sampling.

Block 3: what each $\hat{\epsilon}$ actually measures SUGGESTED

Paths have different Doppler scales: surface bounces add wave-induced Doppler, arrival angles differ. The preamble locks onto the first or the strongest arrival, which can switch over time and differ between the two directions (beam patterns, depths, surface change over τ).

Measurement model for direction $j \in \{A, B\}$:

$$\hat{\epsilon}_j(t_k) = \epsilon(t_k) + b_j(t_k) + w_{j,k}, \qquad w_{j,k} \sim \mathcal{N}(0, R_j)$$

ϵ: common bulk state; b_j: slowly varying bias of the path locked by direction j (a state); R_j: per-direction noise, adapted from the innovations.

Path switches

- innovation gating: a sample far from the prediction is flagged, not fused blindly
- on a flagged jump: reset b_j or inflate its covariance, keep the bulk state
- path-resolved estimates (delay of the locked arrival) as side information, where the receiver gives them

First check OPEN

How often $\hat{\epsilon}$ jumps on Channel Library records, and how far apart are its values on the first and the strongest arrival, before building more machinery.

Block 3: motion sensors and uplink sampling SUGGESTED

Motion sensors, when a node has them

- IMU or GNSS: an extra measurement of the node's *own* motion in the filter
- the projection onto the propagation path is a state, learned from the link Doppler
- link Doppler still gives what the sensor cannot: the far end's motion, the path actually used
- nodes without sensors run on link Doppler alone

Evaluation: link only vs sensors only vs both. This answers the obvious question: why not just an IMU?

Closest: Oiwa et al. 2020, accelerometer-aided compensation at one end.

Uplink with K nodes

- the gateway's periodic beacon gives every node a Doppler sample at once
- the gateway measures each node's uplink packet and returns its value in the next beacon ($\approx$20 bits per node)
- each node runs the two-way filter, with the beacon in place of the ACK

OPEN Beacon length grows with K; can the nodes share the gateway's own motion?

Block 4: pre-compensation at the transmitter SUGGESTED

Why only a forecast can do it. The transmitter's newest $\hat{\epsilon}$ is at least τ old; at 2 km, 2–3 s from beacon to arrival, about a third of an 8 s swell period. Using the last value fails; a forecast over the receive interval does not.

Mechanism (first order): node k sends

$$\tilde{s}_k(t) = s_k\Big(t - \int_0^t \hat{\epsilon}_k \, du\Big)$$

$$\Rightarrow \quad r_k(t) \approx s_k\Big(t + \int_0^t \big(\epsilon_k - \hat{\epsilon}_k\big) du\Big)$$

with $\hat{\epsilon}_k$ aligned to the receive interval: the gateway sees only the residual.

With $K = 1$: an alternative or a complement to Block 1.

What it enables: a concurrent uplink

- K nodes in separate subbands, each with its own Doppler
- at the gateway: one resampling branch per user (Tu et al. 2013), interference remains
- pre-compensated at the source: one FFT, per-user residual tracking

Rough estimate: users 0.45 m/s apart at 20 kHz: $\approx$6 Hz, half a subcarrier; error 0.05 m/s: $\approx$0.7 Hz, interference near -18 dB.

OPEN Only bulk Doppler is removed; path-specific Doppler stays (Zhang 2026).

Decision: a threshold rule on the forecast, written out SUGGESTED

Predicted PER: probability, under the forecast distribution of e over the receive interval, that the residual leaves the calibrated tolerance,

$$\hat{p}_{\mathrm{fail}}(a, x_t) = \Pr_{\hat{d},\sigma_e}\left(e \notin \mathcal{E}_{\mathrm{tol}}(a, \widehat{\mathrm{SNR}})\right)$$

$$\mathcal{S}_t = \left\{a : \hat{p}_{\mathrm{fail}}(a, x_t) \le \alpha - m_t\right\}, \quad a_t = \arg\max_{a \in \mathcal{S}_t} \frac{B_a}{T_a}\left(1 - \hat{p}_{\mathrm{fail}}\right)$$

α: PER target; m_t: margin from delayed ACKs (safety layer). Without the prior ($\hat{d} = 0$): the plain rule predicted drift + margin $\le d_{\mathrm{tol}}$. The PER model is not learned online (follow-up bandit).

The bound must hold over the *whole* receive interval: a pointwise quantile does not bound the maximum.

OPEN $\widehat{\mathrm{SNR}}$ source, bits, age (oracle at first).

Procedure per packet

1. fuse both ends' Doppler, in time order, with gating
2. forecast $\hat{d}$, σ_e per receive interval
3. pilots per action: grid $\mathcal{P}_0$ + extra (Block 2)
4. predicted PER from the calibration curves
5. safe set with m_t; if empty: fallback, wait or drop
6. highest expected goodput in the safe set
7. send frame + forecast; B pre-warps
8. update m_t from delayed ACKs

Uplink: the forecast uncertainty sets guard bands and admission SUGGESTED

Guard band of node k: the offset its residual can reach, at confidence $1-\delta$, over the receive interval,

$$g_k = \bar{f}_k \, z_{1-\delta} \max_u \sigma_{\epsilon,k}(u)$$

$\bar{f}_k$: upper edge of the subband; $\sigma_{\epsilon,k}(u)$: forecast standard deviation of ϵ_k; $z_{1-\delta}$: Gaussian quantile.

Admission and allocation at the gateway:

$$\max_{\mathcal{K},\{S_k\}} \sum_{k\in\mathcal{K}} \frac{B_k}{T}\left(1-\hat{p}_k\right)$$

$$\text{s.t.} \sum_{k\in\mathcal{K}} \left(|S_k| + g_k\right) \le B, \quad \hat{p}_k \le \alpha - m_{k,t}$$

Nodes whose guard band costs more than it is worth are deferred to a later slot.

What this changes

- σ_e gets a measurable effect: how many nodes share the uplink
- the point-to-point rule is the case $K = 1$, $g_1 = 0$
- margin $m_{k,t}$ per node, from its own ACKs

Theory target (R4): inter-user interference vs residuals and guard bands; nodes supported vs range, sea state, PER target.

Closest: Hasan et al. 2025 (no forecast).

OPEN Path-specific Doppler.

Theoretical objectives

Theory: four core results, the rest as extensions or follow-up

R1 Predictability

How long Doppler stays predictable (L1; idea document: $\approx$13 s for an 8 s swell vs $\approx$1 s for fading); mixed seas degrade gracefully (P2)

R2 Sampling

Forecast error vs range, one- vs two-way ($\times 2$ samples per round trip); observability with delayed samples and per-direction bias (L3–L4); gain from motion sensors

R3 Pilots

Optimal extra positions on top of the grid $\mathcal{P}_0$ for one swell component; greedy bound only via weak submodularity (from P7)

R4 Concurrency

Inter-user interference vs forecast residuals and guard bands; how many nodes share the uplink at a given range, sea state and PER target

Weight: R1 and R2 carry the paper, R4 is the main application; R3 if the single-swell case closes.
Follow-up: bandit regret with arm-dependent delay (F1).
Supporting: T10–T11, PER safety layer and its stability (known structure: OLLA / ACI).
Extensions: T8 quiet windows, T9 Doppler vs SNR feedback, L3–L4 path separation, P5, P6.

OPEN Hard parts: path switches and bias in R2; submodularity in R3; path-specific Doppler in R4.
L/P/T: results of the idea document.

Safety layer: what makes the PER claim non-vacuous

Deck claim CLAIM: long-run PER below target for any channel, predictor and estimate.

Margin update. The deck's verbal rule, written as

$$m_{k+1} = m_k + \eta\,(e_k - \alpha), \quad e_k = \mathbf{1}\{\text{failure}\}$$

per processed outcome (α: PER target). Unprojected, $\frac{1}{N}\sum_k e_k = \alpha + \frac{m_{N+1} - m_1}{\eta N}$: calibration only if m grows sublinearly (if all fail, it does not). OPEN Signed correction or buffer $m \geq 0$? (projection changes the proof); outcome vs packet index.

Which guarantee? Long-run frequency, finite-horizon, or per-packet. Report it with goodput, drops, deadline misses, retransmissions.

To fix before proving Thm. 10 OPEN

- PER denominator: packets sent? with retransmissions? deadline expiries?
- missing ACK or timeout: failure, or excluded?
- empty feasible set: fallback action, wait, or drop, and how each is counted
- minimum service required, e.g. a bound on the dropped or delayed fraction
- feasibility assumption, e.g. some action always meets the target with slack
- action-dependent delay: outcomes arrive $\geq h + 2\tau + T_a$ late

Risks found by the prior-art search

R3: no free greedy bound. Kalman sensor-scheduling costs are not submodular in general (Jawaid–Smith, Automatica 2015); the Krause et al. 2008 bound covers Gaussian-process placement only. Routes: closed form for one swell; submodularity ratio or curvature (Hashemi et al., TAC 2020); Jawaid–Smith conditions for our model.

Block 2: periodic is optimal when stationary. Uniform pilots minimise steady-state MSE for a Gauss–Markov channel (Dong–Tong–Sadler 2002). Extras pay off only where $\sigma_e^2(u)$ varies within the packet: state it as a precondition.

Block 1: field counter-evidence. Surface-buoy trial, 10.5 h, 26 170 frames: per-symbol Doppler refinement lowers PER by at most 1.15× (Lee, Park, Kim, Im, SSRN 2026). Block 1 must be shown in a regime fixed in advance (long packets, strong swell, few pilots), against acceleration-aware baselines (Zakharov 2019, Cho–Kim 2026, Hu 2024).

Framing. DE-Sync's exchange, navigation filters on modem Doppler and NTN make "close the loop" look obvious. Defence: the analysis (R1 horizon, R2 two-way sampling with bias, R3 placement, R4 concurrency), not the plumbing.

Follow-up

Follow-up paper: a contextual bandit that knows the physics SUGGESTED

Not in the JSAC submission: it replaces the threshold rule once the calibration curves are learned online.

Context x_t: forecast over the candidate receive interval ($\hat{d}$, σ_e), $\widehat{\mathrm{SNR}}$, range.

Arms $a = (M, N_{\mathrm{sym}})$; for each arm, $\mathcal{P}$ comes from Block 2.

Reward: delivered bits per unit airtime, $B_a \mathbf{1}\{\text{success}\}/T_a$ (one objective, fixed before the comparison).

Model: failure probability monotone in forecast error and action; generalised linear in physical features,

$$p_{\mathrm{fail}}(a, x) = g\big(\theta_a^\top \psi(x)\big),$$

θ_a initialised from the calibration curves, learned online (UCB or Thompson sampling).

What is new for the theory (F1)

- feedback for arm a arrives $D_a \geq h + 2\tau + T_a$ later: the delay depends on the arm
- short packets return feedback sooner, biasing a naive learner towards them OPEN
- target CLAIM: regret $\tilde{O}(\sqrt{T})$ + delay term, extending fixed-delay results (Joulani et al., ICML 2013)

Safety layer: only arms meeting the PER target with margin m_t.

Builds on AMUSE, POSEIDON, AoI bandit (Busacca et al.). Details: `followup_bandit.md`.

Questions for the group

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- Bandit moved to a follow-up: is a calibrated threshold rule enough for the JSAC paper?
- Which receiver structure sets $K_{\max}$: comb pilots in every symbol, or full training symbols only?
- Can the target modem or simulator carry $\approx$20 header bits for the forecast and the far end's Doppler, and K such fields in the beacon?
- What is already implemented, and which results are proved?
- May the policy abstain, drop packets or use a fallback, and how is service counted?
- How much time and theory/simulation support is available?
- Target uplink: buoy gateway with seabed nodes, AUVs, or both? Which nodes have an IMU or GNSS?
- First test: does a forecast prior still help acceleration-aware receivers, in PER or in pilots saved?